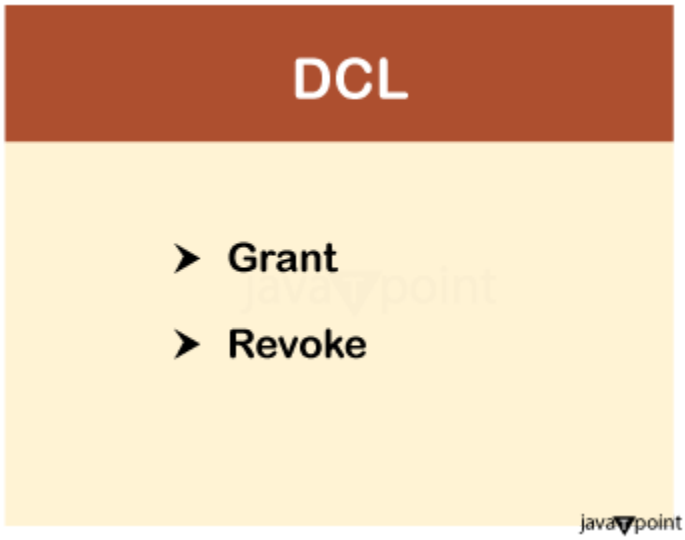
Day-6

3. Data Control Language

DCL commands are used to grant and take back authority from any database user.

Following are the some commands that come under DCL:



Syntax:

```
GRANT <obj_priv> ON <obj_name> To <username>;
```

In the above syntax, obj_priv> is the DML statement like Insert, Delete , update and Select and <obj_name> is a table, view etc. and username is the name of the authorized user.

Example

```
GRANT SELECT, UPDATE ON MY_TABLE TO SOME_USER,
ANOTHER_USER;
```

b. Revoke: It is used to take back permissions from the user.

Syntax:

```
REVOKE <obj_priv> ON <obj_name> FROM <username>;
```

In the above syntax, obj_priv> is the DML statement like Insert, Delete , update and Select and <obj_name> is a table, view etc. and username is the name of the user from whom the permission is revoked.

Example

```
REVOKE SELECT, UPDATE ON MY_TABLE FROM USER1, USER2;
```

Grant and Revoke

```
==========================
mysql> use codegnan;
Database changed
mysql> show tables;
+-------------------+
| Tables_in_codegnan |
+-------------------+
| codegnan_emp       |
+-------------------+
1 row in set (0.23 sec)

mysql> create table students(
    -> name varchar(20),
    -> age int,
    -> mobile varchar(12)
    -> );
Query OK, 0 rows affected (0.15 sec)

mysql> insert into students values('student 1',23,'9182454984');
Query OK, 1 row affected (0.02 sec)

mysql> select*from students;
+-----------+------+------------+
| name      | age  | mobile     |
+-----------+------+------------+
| student 1 |   23 | 9182454984 |
+-----------+------+------------+
1 row in set (0.00 sec)

mysql> create user user1 identified by 'user1';
Query OK, 0 rows affected (0.10 sec)

mysql> grant select,insert,update on students to user1;
Query OK, 0 rows affected (0.01 sec)

mysql> create user user2 identified by 'user2';
```

Query OK, 0 rows affected (0.01 sec)

mysql> grant select,update on students to user2;
Query OK, 0 rows affected (0.01 sec)

user1 previligeous

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u USER1 -p
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 75
Server version: 8.0.39 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use codegnan;
Database changed
mysql> select*from user1;
ERROR 1142 (42000): SELECT command denied to user 'user1'@'localhost' for table 'user1'
mysql> select*from students;
+-----------+------+------------+
| name | age | mobile |
+-----------+------+------------+
| student 1 | 23 | 9182454984 |
+-----------+------+------------+
1 row in set (0.00 sec)

mysql> insert into students values('student 2',32,'8834248768');
Query OK, 1 row affected (0.01 sec)

mysql> insert into student(name,mobile) values('student 3','874772372'),('student 4','689648797');

ERROR 1142 (42000): INSERT command denied to user 'user1'@'localhost' for table 'student'
mysql> insert into students(name,mobile) values('student 3','874772372'),('student 4','689648797');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> update students set age=37 where name='student4';
Query OK, 0 rows affected (0.01 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> drop table student;
ERROR 1142 (42000): DROP command denied to user 'user1'@'localhost' for table 'student'
mysql> delete table students;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'table students' at line 1
mysql> delete from students where age=37;;
ERROR 1142 (42000): DELETE command denied to user 'user1'@'localhost' for table 'students'
ERROR:
No query specified

mysql>

user2 previloges
==========
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u USER2 -p
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 79
Server version: 8.0.39 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> select*from students;
ERROR 1046 (3D000): No database selected
mysql> use codegnan;
Database changed
mysql> select*from students;
+-----------+------+------------+
| name      | age  | mobile     |
+-----------+------+------------+
| student 1 |   23 | 9182454984 |
| student 2 |   32 | 8834248768 |
| student 3 | NULL | 874772372  |
| student 4 | NULL | 689648797  |
+-----------+------+------------+
4 rows in set (0.01 sec)

mysql> insert into students values('djs',09,'9889');
ERROR 1142 (42000): INSERT command denied to user 'user2'@'localhost' for
table 'students'
mysql> drop table students;
ERROR 1142 (42000): DROP command denied to user 'user2'@'localhost' for
table 'students'
mysql> update students age=40 where name='student 4';
ERROR 1064 (42000): You have an error in your SQL syntax; check the
manual that corresponds to your MySQL server version for the right
syntax to use near '=40 where name='student 4'' at line 1
mysql> update students set age=40 where name='student 4';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select*from students;
+-----------+------+------------+
| name      | age  | mobile     |
+-----------+------+------------+
| student 1 |   23 | 9182454984 |
| student 2 |   32 | 8834248768 |
| student 3 | NULL | 874772372  |
| student 4 |   40 | 689648797  |
+-----------+------+------------+
4 rows in set (0.00 sec)

mysql>
```

==========
===
Revoke
---
mysql> revoke update on students from user2;
Query OK, 0 rows affected (0.01 sec)

update students set age=40 where name='student 4';
ERROR 1142 (42000): UPDATE command denied to user 'user2'@'localhost' for table 'students'

```
CREATE TABLE EMPLOYEES(
EMPID INT AUTO_INCREMENT PRIMARY KEY,
NAME VARCHAR(100),
DEPARTMENT VARCHAR(50)
);

INSERT INTO EMPLOYEES(NAME,DEPARTMENT)
VALUES
('MALLI','IT'),
('INDU','HR');

SELECT*FROM EMPLOYEES;
| EMPID | NAME  | DEPARTMENT |
|     1 | MALLI | IT         |
|     2 | INDU  | HR         |

CREATE TABLE BACKUPEMPLOYEES AS
SELECT*FROM EMPLOYEES;

DESC BACKUPEMPLOYEES;
| Field      | Type         | Null | Key | Default | Extra |
| EMPID      | int          | NO   |     | 0       |       |
| NAME       | varchar(100) | YES  |     | NULL    |       |
| DEPARTMENT | varchar(50)  | YES  |     | NULL    |       |
```

```
SELECT*FROM BACKUPEMPLOYEES;
| EMPID | NAME  | DEPARTMENT |
|     1 | MALLI | IT         |
|     2 | INDU  | HR         |

CREATE TABLE PRODUCTS(
PRODUCTID INT PRIMARY KEY,
PRICE DECIMAL(10,2),
QUANTITY INT,
TOTALAMOUNT DECIMAL(10,2) GENERATED ALWAYS AS(PRICE*QUANTITY)
STORED
);

INSERT INTO PRODUCTS(PRODUCTID,PRICE,QUANTITY)
VALUES
(1,50,3),
(2,100.25,6);

SELECT*FROM PRODUCTS;
| PRODUCTID | PRICE  | QUANTITY | TOTALAMOUNT |
|         1 |  50.00 |        3 |      150.00 |
|         2 | 100.25 |        6 |      601.50 |

UPDATE PRODUCTS SET PRICE=500 WHERE PRODUCTID=1;

SELECT*FROM PRODUCTS;
| PRODUCTID | PRICE  | QUANTITY | TOTALAMOUNT |
|         1 | 500.00 |        3 |     1500.00 |
|         2 | 100.25 |        6 |      601.50 |
```

DDL,DML,DQL practice Questions and answers

==========

Table creation

==========

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(100),
    department VARCHAR(50),
    job_role VARCHAR(50),
    salary DECIMAL(10,2),
    joining_date DATE,
```

```
    city VARCHAR(50)
);

INSERT INTO employees
(emp_id, emp_name, department, job_role, salary, joining_date, city)
VALUES
(101, 'Ravi Kumar', 'IT', 'Developer', 45000, '2023-01-15', 'Vizag'),
(102, 'Priya Sharma', 'HR', 'HR Manager', 55000, '2022-06-20', 'Hyderabad'),
(103, 'Kiran Reddy', 'IT', 'Tester', 40000, '2023-03-10', 'Vizag'),
(104, 'Sneha Rao', 'Finance', 'Accountant', 48000, '2021-11-25', 'Chennai'),
(105, 'Arun Kumar', 'IT', 'Developer', 60000, '2020-08-18', 'Bangalore'),
(106, 'Divya Reddy', 'HR', 'Recruiter', 35000, '2024-01-05', 'Vizag'),
(107, 'Suresh Babu', 'Sales', 'Executive', 42000, '2022-09-12', 'Hyderabad'),
(108, 'Anjali Devi', 'IT', 'Developer', 70000, '2019-05-30', 'Chennai'),
(109, 'Mahesh Kumar', 'Sales', 'Manager', 65000, '2020-02-14', 'Bangalore'),
(110, 'Lakshmi Rao', 'Finance', 'Analyst', 52000, '2023-07-22', 'Hyderabad');


Update Examples
==================
-- 1. Update the salary of Ravi Kumar
UPDATE employees
SET salary = 50000
WHERE emp_id = 101;

-- 2. Increase salary by 10% for all IT employees
UPDATE employees
SET salary = salary * 1.10
WHERE department = 'IT';

-- 3. Change Priya Sharma's city
UPDATE employees
SET city = 'Vizag'
WHERE emp_id = 102;

-- 4. Change Kiran Reddy's job role
UPDATE employees
SET job_role = 'Senior Tester'
WHERE emp_id = 103;

-- 5. Increase salary by 5000 for all HR employees
UPDATE employees
```

```
SET salary = salary + 5000
WHERE department = 'HR';

-- 6. Change all employees in Vizag to Hyderabad
UPDATE employees
SET city = 'Hyderabad'
WHERE city = 'Vizag';

-- 7. Change Arun Kumar's department
UPDATE employees
SET department = 'Development'
WHERE emp_id = 105;

-- 8. Increase salary by 15% for employees earning less than 45000
UPDATE employees
SET salary = salary * 1.15
WHERE salary < 45000;

-- 9. Change Anjali Devi's job role
UPDATE employees
SET job_role = 'Senior Developer'
WHERE emp_id = 108;

-- 10. Update the city of all Finance employees to Bangalore
UPDATE employees
SET city = 'Bangalore'
WHERE department = 'Finance';

Delete Examples
===============
-- 1. Delete employee with ID 106
DELETE FROM employees
WHERE emp_id = 106;

-- 2. Delete employees from the Finance department
DELETE FROM employees
WHERE department = 'Finance';

-- 3. Delete employees earning less than 40000
DELETE FROM employees
WHERE salary < 40000;
```

```sql
-- 4. Delete employees working in Vizag
DELETE FROM employees
WHERE city = 'Vizag';

-- 5. Delete employee named Suresh Babu
DELETE FROM employees
WHERE emp_name = 'Suresh Babu';

-- 6. Delete employees who joined before 2020
DELETE FROM employees
WHERE joining_date < '2020-01-01';

-- 7. Delete employees working as Recruiter
DELETE FROM employees
WHERE job_role = 'Recruiter';

-- 8. Delete employees from the Sales department earning less than 50000
DELETE FROM employees
WHERE department = 'Sales'
AND salary < 50000;

-- 9. Delete employee with the lowest salary
DELETE FROM employees
WHERE salary = (
    SELECT MIN(salary)
    FROM employees
);

-- 10. Delete employees whose city is Chennai
DELETE FROM employees
WHERE city = 'Chennai';
```

Select Queries
==========

```sql
-- 1. Display all employees
SELECT * FROM employees;

-- 2. Display employees working in IT
SELECT *
FROM employees
WHERE department = 'IT';
```

```sql
-- 3. Display employees earning more than 50000
SELECT *
FROM employees
WHERE salary > 50000;

-- 4. Display employee names and salaries
SELECT emp_name, salary
FROM employees;

-- 5. Display employees from Vizag
SELECT *
FROM employees
WHERE city = 'Vizag';

-- 6. Display employees who joined after 2022-01-01
SELECT *
FROM employees
WHERE joining_date > '2022-01-01';

-- 7. Display employees earning between 40000 and 60000
SELECT *
FROM employees
WHERE salary BETWEEN 40000 AND 60000;

-- 8. Display employees whose name ends with 'Kumar'
SELECT *
FROM employees
WHERE emp_name LIKE '%Kumar';

-- 9. Display employees in descending order of salary
SELECT *
FROM employees
ORDER BY salary DESC;

-- 10. Display the highest-paid employee
SELECT *
FROM employees
ORDER BY salary DESC
LIMIT 1;
```